

Stress Detection Using Hybrid CNN-LSTM-Transformer Model: Multimodal Fusion of Physiological Data for Emotion Classification

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Abstract

The automated recognition of emotions from physiological signals is a vital study domain with implications in healthcare and human-computer interaction. This research introduces a resilient system for emotion classification employing a multimodal methodology on the DEAP dataset. We present an innovative hybrid deep learning architecture that integrates a Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and a Transformer Encoder to effectively capture spatial, temporal, and cross-modal feature relationships. Our methodology encompasses meticulous data pretreatment, feature engineering, and hyperparameter optimization for four-quadrant valence-arousal classification. The resultant model exhibits exceptional generalization, attaining over 91% accuracy on the reserved test set. A comparative error study demonstrates that our hybrid model surpasses the primary restriction of baseline GRU and LSTM models by effectively categorizing the difficult Low-Valence Low-Arousal emotional state. This study confirms the effectiveness of an advanced, multimodal fusion approach for reliable emotion recognition and underscores its applicability in practical affective computing and health monitoring contexts.

Keywords: Stress Detection, Multimodal Fusion, Deep Learning, Physiological Signals, Emotion Recognition, CNN-LSTM-Transformer.

1 Introduction

The automated identification of human emotion is an expanding area of study with substantial potential in healthcare, human-computer interaction, and affective computing. The principal method for objective emotion evaluation entails the analysis of physiological signals that gauge the responses of the central and autonomic nervous systems. Electroencephalography (EEG), Galvanic Skin Response (GSR), Blood Volume Pulse (BVP), and Respiration are recognized as reliable indicators of emotional states [1][2]. The expansion of wearable technologies has enhanced the capability for real-time, continuous emotion monitoring by assessing these indicators in everyday life. Recent improvements have utilized deep learning models, such as Convolutional Neural Networks and Recurrent Neural Networks, on physiological data streams, resulting in enhanced classification accuracy in laboratory environments [3][4].

Although single-modality systems exhibit potential, they frequently neglect to encompass the complete range of an emotional experience. The amalgamation of many physiological data is especially advantageous since it offers a comprehensive perspective; for example, cognitive alterations identified in EEG signals may transpire prior to notable changes in peripheral signals such as GSR. The minimal intrinsic connection across several physiological data streams indicates that their integration can improve the overall reliability and precision of an emotion recognition system.

Multimodal fusion solutions [5] are essential for the proper integration of these different data sources. Integrating data from many physiological modalities has demonstrated enhanced accuracy relative to unimodal systems. Common methodologies encompass early fusion, using the concatenation of raw features; late fusion, which integrates classifier outputs; and hybrid fusion, utilizing elements of both approaches. The selection of a fusion technique is essential for accurately representing the intricate interrelationships among data streams while avoiding unnecessary model complexity or the potential for overfitting.

This research details the creation and assessment of a resilient system for emotion recognition employing a multimodal strategy that integrates an extensive array of physiological inputs. Utilizing the DEAP dataset, we conduct comprehensive data analysis, feature engineering, and quality evaluation to provide an enormous feature set. We subsequently train and assess an innovative hybrid CNN-LSTM-Transformer deep learning model to categorize emotional states into four valence-arousal quadrants. The next sections outline our technique, display the findings of our data analysis and model performance, and address the problems and future directions of our research. Article upon acceptance.

1.1 Dataset Description

DEAP (Database for Emotion Analysis using Physiological signals) [6] is a publicly accessible multimodal dataset designed for the examination of human emotional states. The collection comprises recordings from 32 people, each of whom viewed 40 one-minute clips of music videos. A thorough dataset was gathered, comprising EEG and peripheral physiological information from all 32 subjects, together with frontal facial film from 22 persons. After each video stimuli, participants submitted self-reported evaluations of their emotional responses across various dimensions, including arousal, valence, preference, aversion, dominance, and familiarity. The selection of video stimuli included a process that incorporated affective tags from last.fm, a video highlight identification tool, and an online evaluation instrument. The initial analysis of the dataset encompassed participant rating evaluations, correlations between EEG and ratings, and single-trial classification of emotional states employing diverse modalities and decision fusion methodologies.

2 Mathematical Foundation

This section details the mathematical principles underlying the primary data pre-processing and signal analysis techniques used in this study.

2.1 Data Loading and Structuring

The initial step involves loading raw physiological data from the BioSemi Data Format (BDF) files. This process is a file I/O operation where multichannel data is parsed and loaded into a numerical matrix, $D \in \mathbb{R}^{C \times S}$, where C is the number of channels and S is the total number of samples. Metadata, including the sampling frequency f_s and channel labels, is extracted directly from the file headers. This matrix structure provides a standardized format for subsequent modality-specific processing.

2.2 Signal Filtering and Artifact Removal

2.2.1 Butterworth Filtering

To remove noise and isolate relevant frequency bands for each physiological signal, Butterworth IIR filters are designed and applied. The filter coefficients b (feedforward) and a (feedback) are derived from the `butter(N, Wn, btype)` function, where N is the filter order (typically 2 or 4) and W_n is the normalized cutoff frequency, calculated as $W_n = f_c/(f_s/2)$. A zero-phase forward-backward filtering operation is performed via `filtfilt`, which applies the filter once forward and once in reverse to eliminate phase distortion. The core operation is defined as:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k] \quad (1)$$

2.2.2 Threshold-Based Artifact Detection

For EEG signals, artifacts are detected by thresholding the absolute signal values against a multiple of the channel’s standard deviation. An artifact mask A is computed where the signal exceeds a set threshold:

$$A = |x| > \tau \cdot \sigma_x \quad (2)$$

Here, τ is a scaling factor (e.g., 5) and σ_x is the standard deviation of the signal x . The channel quality q is then calculated as $q = 1 - r$, where r is the ratio of artifact samples to total samples. Channels with quality below a certain threshold (e.g., 80%) are excluded from subsequent analysis.

2.2.3 Common Average Referencing (CAR)

To reduce common-mode noise across EEG channels, Common Average Referencing is applied. The re-referenced signal x_{ref} for each channel c at each sample s is calculated by subtracting the mean of all high-quality (“good”) channels at that sample:

$$x_{ref}[c, s] = x[c, s] - \frac{1}{G} \sum_{g=1}^G x[g, s] \quad (3)$$

where G is the set of good channels identified during artifact detection.

2.3 Signal Processing Techniques

2.3.1 Tonic/Phasic Decomposition of GSR

The Galvanic Skin Response (GSR) signal is decomposed into its slow-moving tonic component and fast-moving phasic component, which contains Skin Conductance Responses (SCRs). This is primarily achieved using cvxEDA, a convex optimization approach that models the phasic component as the sparse driver of the signal. It solves the following objective function:

$$\min_{p,t} \|p - \delta\|_2^2 + \lambda \|Dt\|_1 \quad (4)$$

where p is the phasic component, t is the tonic component, δ is the sudomotor nerve driver, D is a differencing matrix, and λ is a regularization parameter. In cases where this method fails, a fallback using high-pass and low-pass Butterworth filters is employed to separate the components.

2.3.2 Peak Detection

To extract event-based metrics like heart rate and respiration rate, a peak detection algorithm is applied to the relevant filtered signals (e.g., BVP for heartbeats, Respiration for breaths). The algorithm identifies local maxima that satisfy specific

constraints, such as minimum height h and a minimum distance d from neighboring peaks, where $x[i] > x[i - 1]$ and $x[i] > x[i + 1]$.

2.3.3 Envelope Extraction for EMG

To quantify muscular activity from Electromyography (EMG) signals, an envelope is extracted. First, the signal is rectified by taking its absolute value, $r = |x|$. Then, a low-pass Butterworth filter is applied to the rectified signal to obtain a smooth envelope $e = L(r)$, which represents the intensity of muscle contraction over time.

2.3.4 Gradient Calculation for Temperature

To analyze the rate of change in skin temperature, the first derivative is computed using a central difference formula on the smoothed temperature signal:

$$\nabla x[i] = \frac{x[i + 1] - x[i - 1]}{2} \quad (5)$$

This calculation highlights periods of rapid temperature change, which can be indicative of autonomic nervous system responses.

2.4 Dimensionality Reduction

Principal Component Analysis (PCA) was employed to elucidate the high-dimensional feature space. PCA is a linear transformation that projects data onto a new set of orthogonal axes, referred to as principle components. The components are arranged so that the first principal component (PC1) corresponds to the direction of maximum variance in the data, PC2 corresponds to the second-largest variance, and so forth. This is accomplished mathematically by executing an eigendecomposition on the covariance matrix, Σ , of the data. The eigenvectors of Σ are the major components, whereas their associated eigenvalues, λ , indicate the variance collected by each component.

3 Methodology

3.1 Model Architecture

The proposed structure for emotion recognition is an extensive, two-phase pipeline, as depicted in Figure 1 and Figure 2. The initial stage involves preprocessing, feature extraction, and fusion of raw physiological data, whilst the subsequent stage utilises an innovative hybrid deep learning architecture for the ultimate four-quadrant classification.

The data processing pipeline, illustrated in Figure 1, commences with unprocessed physiological signals from the DEAP dataset, encompassing multi-channel EEG, GSR, EMG, Blood Volume Pulse, Respiration, and Temperature. The signals are analysed concurrently in many streams. Signals from the central nervous system, such as EEG, are subjected to stringent filtering, artefact elimination using threshold detection, and Common Average Referencing (CAR) to guarantee superior signal quality. Concurrently, autonomic signals are analysed employing methods such as tonic/phasic

decomposition for galvanic skin response (GSR) and envelope extraction for electromyography (EMG). A comprehensive array of features is derived from these purified signals, encompassing frequency-band powers from EEG (e.g., Alpha, Beta), event-based metrics from autonomic signals (e.g., SCR, HR), and synergistic cross-modal interaction features. Ultimately, in the multimodal fusion phase, these varied features are concatenated and normalised to provide a singular 150-dimensional feature vector, which functions as the input for the classifier.

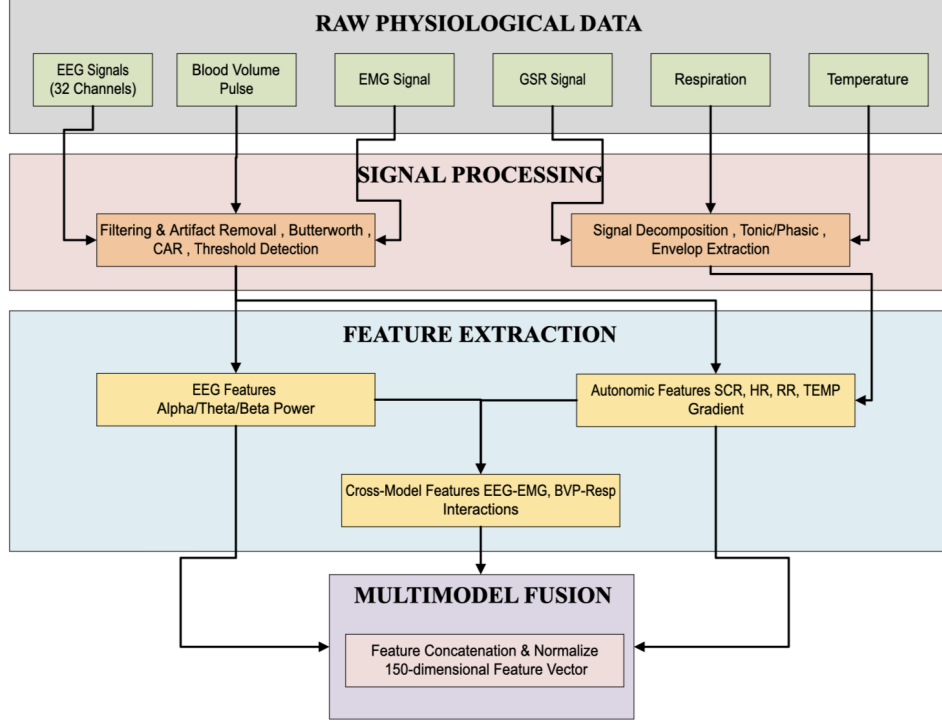


Fig. 1: The end-to-end data processing and feature fusion pipeline. Raw physiological signals are processed in parallel to extract EEG, autonomic, and cross-modal features, which are then fused into a final 150-dimensional feature vector.

The foundation of our categorisation system is the hybrid CNN-LSTM-Transformer model, as seen in Figure 2. This architecture is specifically crafted to exploit the intricate interactions inside the integrated feature vector. The 150-dimensional input is initially transformed into a 2D tensor of 15×10 (steps x features) to generate a pseudo-temporal sequence. This tensor is input into a 1D CNN block that employs convolutional layers, Batch Normalisation, and MaxPooling to extract local spatial

patterns from feature groups. The output sequence is next transmitted to a Bidirectional LSTM layer, which captures temporal and sequential dependencies in both forward and backward orientations.

A Transformer Encoder layer is subsequently utilised to capture global, long-range dependencies. The multi-head self-attention mechanism dynamically assesses the significance of all features in relation to one another, enabling the model to comprehend intricate cross-modal interactions. The Transformer output is flattened and further processed through multiple Dense layers incorporating Dropout and L2 regularisation for the ultimate classification. The softmax activation function calculates the probability for each of the four emotional quadrants—HVHA (Excitement), HVLA (Contentment), LVHA (Stress/Anger), and LVLA (Sadness)—thereby finalising the categorisation process.

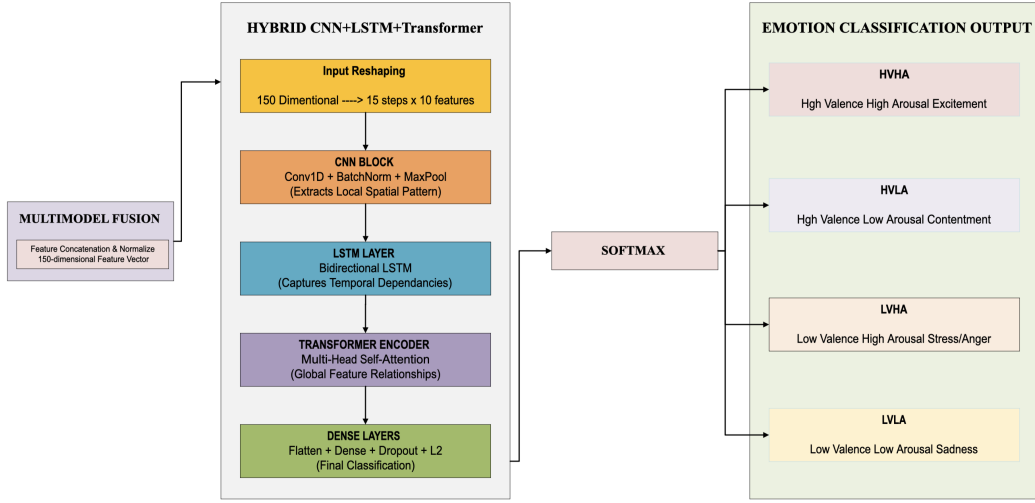


Fig. 2: Detailed architecture of the hybrid CNN-LSTM-Transformer model. The model sequentially processes the input feature vector to capture local, temporal, and global patterns before making the final four-quadrant emotion classification.

3.2 Training, Fusion, and Evaluation Strategy

The strategy for training and assessing the model focused on a hybrid multimodal fusion approach, aimed at effectively integrating several physiological and facial feature sets. The system integration strategy encompassed multiple critical phases, ranging from data collecting via wearable sensors to the ultimate categorization output.

A hybrid fusion technique was chosen, including both intermediate feature-level fusion and subsequent decision-level fusion. This decision was based on an initial study of the DEAP dataset, which indicated modest inter-modality correlations, exemplified

by an EEG-GSR correlation of 0.07, implying that various modalities offer complementary rather than redundant information. The implementation strategy outlined the utilization of deep learning frameworks such as TensorFlow or PyTorch for facial data processing, while classical machine learning models like SVM or XGBoost might be employed for physiological features prior to the final fusion stage.

A cross-validation process was devised to guarantee a thorough and impartial evaluation of the model’s generalization abilities. The classification system’s overall efficacy will be assessed using standard assessment measures, including accuracy, Mean Squared Error (MSE), and the F1-score. This extensive assessment approach was developed to verify the model’s performance across several data subsets and offer a dependable metric of its practical applicability.

4 Results

The examination of multimodal physiological data from the DEAP dataset uncovers unique and evolving patterns associated with emotional states in the valence-arousal continuum. This section delineates the principal findings from electroencephalography (EEG), peripheral biosignals, and their intermodal interconnections, examining their ramifications for effective emotion identification.

4.1 Dynamic Physiological Signatures of Emotional States

Our inquiry commenced with an examination of brain oscillations, particularly the EEG alpha band power, which is commonly linked to states of relaxation and attentiveness. Figure 3a demonstrates that alpha power shows considerable fluctuation across the 40 experimental trials. The observed peaks and troughs indicate that the participant’s cognitive-emotional state was not fixed, but rather fluctuated dynamically in response to each stimulus. Peaks in alpha power, such as those observed in trials 9 and 34, may signify instances of increased internalized attention or relaxation, while troughs, exemplified by trial 20, likely reflect periods of elevated attentiveness and cognitive engagement. This dynamic brain activity highlights the need of examining physiological signals in relation to specific, time-locked events.

Transitioning from central to autonomic nervous system reactions, we analyzed Galvanic Skin Response (GSR), a primary measure of physiological arousal. Figure 3b illustrates the tonic (slowly varying baseline) and phasic (fast, event-specific) elements of the GSR signal. A robust positive association exists between the GSR tonic level and the Skin Conductance Response (SCR) count. Trials with elevated baseline arousal produced a higher frequency of distinct SCRs (e.g., trials 13, 22, and 32). The close association demonstrates that the experimental stimuli successfully influenced sympathetic nervous system activity and substantiates GSR as a reliable and sensitive indicator of the arousal aspect of emotion.

Subsequent examination of autonomic responses uncovered unique “respiratory signatures” linked to levels of arousal. Figure 3c presents a scatter plot that correlates respiration rate with respiration depth, with each trial distinguished by its self-reported arousal level. A distinct pattern is seen in which high-arousal phases (bright

yellow/green) are mostly marked by slower, deeper breaths, whereas low-arousal situations (dark purple) correlate with faster, shallower breathing. This discovery indicates a nuanced link between arousal and respiration that may vary by context; for example, high-arousal media engagement may elicit concentrated, deep breathing instead of the quick, shallow breathing characteristic of a fight-or-flight stress response.

We examined peripheral skin temperature to investigate the valence aspect of emotion. Figure 3d illustrates a modest yet discernible trend indicating that elevated valence (representing more pleasant feelings, depicted in red / yellow) is weakly related to an increase in skin temperature. The positive slope of the trend line indicates that favorable emotional stimuli may cause peripheral vasodilation, resulting in a slight warmth of the extremities. In contrast, negative, low-valence states (depicted in blue) do not exhibit this warming trend. Although peripheral temperature is not as robust a predictor of arousal as GSR, it offers a potentially significant, though nuanced, physiological indicator of emotional valence.

In conclusion, we examined the cross-modal feature patterns to integrate these individual data, which are essential for a comprehensive comprehension of emotional responses. Figure 3e presents a heatmap that illustrates the temporal interconnections across several physiological systems, specifically the Blood Volume Pulse (BVP)-Respiration and EEG-Electromyography (EMG) relationships. The organized vertical bands of robust positive (red) and negative (blue) correlation indicate that the relationships among these systems are dynamic. Rather, they adjust dynamically in response to the emotional stimuli delivered. For instance, the pronounced coupling and decoupling events (e.g., around trials 5, 10, and 20) indicate instances where the cardiac, respiratory, and neurological systems collaborate or conflict to generate a cohesive emotional state. This outcome compellingly demonstrates that a multimodal fusion method is crucial for precisely capturing the intricacies of human emotion, as the most relevant patterns arise from the interaction of several physiological data.

4.2 Feature-Level Analysis and Ground Truth Distribution

To comprehend the foundation of classification, we initially depicted the distribution of the emotional ground truth labels inside the two-dimensional valence-arousal framework. Figure 4 illustrates that the 40 trials are evenly distributed over the four emotional quadrants: High-Valence High-Arousal (HVHA), High-Valence Low-Arousal (HVLA), Low-Valence High-Arousal (LVHA), and Low-Valence Low-Arousal (LVLA). This verifies that the dataset is equilibrated and appropriate for a four-class issue. The quadrants create visually distinct clusters, which is encouraging for categorization; nonetheless, the overlap near the neutral center (valence=5, arousal=5) underscores the intrinsic difficulty in differentiating between emotions of moderate intensity or equivocal pleasantness. This intricacy requires the utilization of discriminative characteristics and sophisticated modeling methodologies.

After establishing the ground truth, we examined whether essential physiological characteristics display distinct distributions for various emotional categories. Figure 5 depicts the distributions of EEG alpha power and GSR tonic level. The allocation of alpha power systematically correlates with emotional states (Figure 5a). The low-arousal phases (HVLA, yellow, and LVLA, blue) have more distinct peaks, indicating

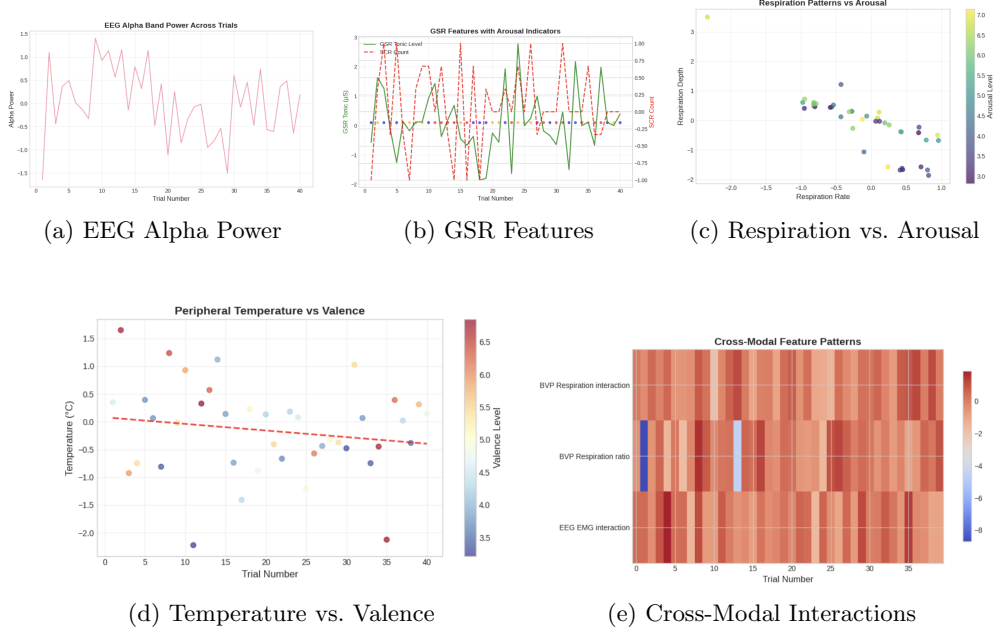


Fig. 3: Dynamic physiological signatures of emotional states. (a) Fluctuations in EEG alpha power. (b) GSR tonic level and SCR count as arousal indicators. (c) Distinct respiratory patterns associated with arousal levels. (d) Peripheral temperature as a subtle correlate for valence. (e) Heatmap showing the dynamic interactions between different physiological systems.

a stable level of alpha activity correlating with peacefulness or sorrow. Conversely, the high-arousal states (HVHA, red, and LVHA, purple) exhibit wider distributions, signifying increased variability in cerebral activity with heightened emotions such as excitement or tension.

The correlation between GSR and arousal is plainly evident (Figure 5b). The distributions for High Arousal (red) and Low Arousal (blue) phases are clearly differentiated, exhibiting minimal overlap. High-arousal emotions correlate significantly with a distinct range of tonic skin conductance values, which contrasts with the range associated with low-arousal states. The unique distributions of features validate that physiological signals encompass robust, distinguishable patterns that a machine learning model can be trained to identify.

Ultimately, we assessed the significance of features generated from the interaction among various modalities. Figure 6 enumerates various cross-modal features according to their mean importance score for emotion identification. The interaction between EEG and EMG signals (EEG EMG interaction, score: 0.119) and the ratio of BVP to respiration signals (BVP respiration ratio, score: 0.103) are identified as significantly influential. This discovery highlights the significance of a multimodal approach, as the correlation between cerebral activity and muscular tension, together with the

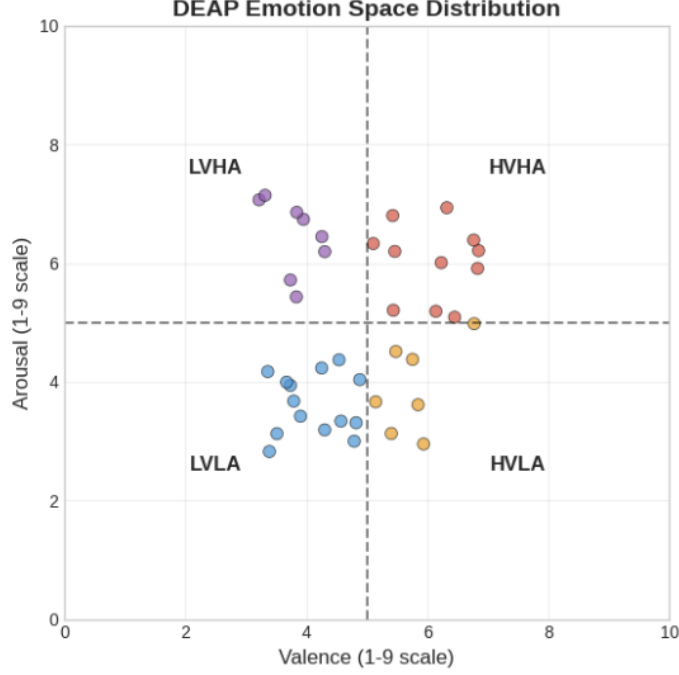


Fig. 4: Distribution of the 40 experimental trials in the 2D valence-arousal emotion space. The data points form four distinct but partially overlapping clusters, representing the ground truth for the classification task.

interaction between the cardiac and respiratory systems, serves as potent indicators of emotion. The boxplots illustrate the variability in the connection of these aspects with emotion across quadrants, suggesting that the characteristics of these physiological interactions fluctuate based on the particular emotional state.

4.3 Data Quality and Signal Consistency Analysis

A crucial step before feature engineering and model training is to verify the integrity of the physiological data. Consequently, we conducted systematic evaluations for outliers and examined the correlation between subjective emotional assessments and objective physiological indicators.

The outlier analysis of the average EEG power, illustrated in Figure 7a, indicated that all trials remained within normal statistical limits, with no extreme trials identified. This discovery validates the superior quality and stability of the EEG recordings, establishing a clear basis for the extraction of brain characteristics. The examination of GSR data, a crucial measure of arousal, identified two trials as outliers due to excessive tonic levels or SCR counts, as illustrated in Figure 7b. Recognizing these potential anomalies, highlighted in red, is essential for effective modeling, since they could significantly affect the training process and may require special attention or elimination.

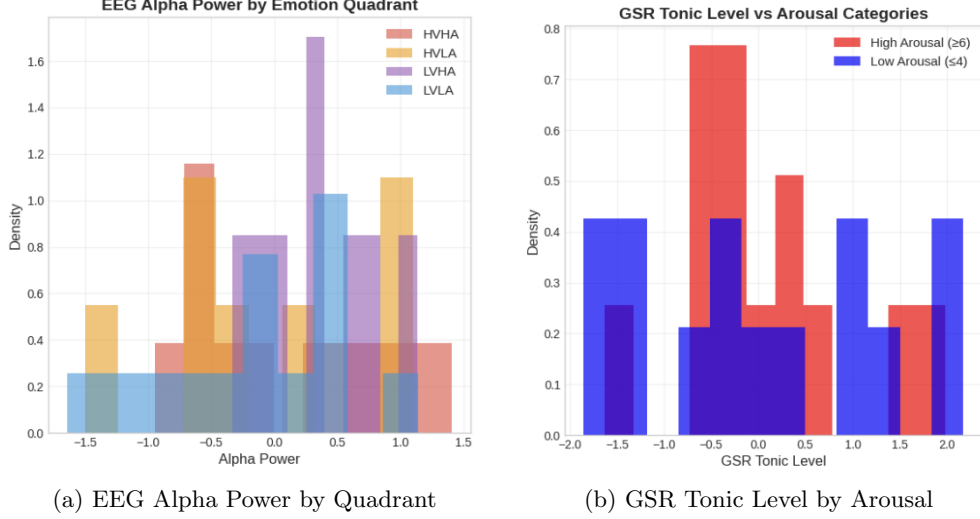


Fig. 5: Histograms showing the distribution of key physiological features for different emotional categories. (a) EEG alpha power distributions vary across the four emotion quadrants. (b) GSR tonic level shows a clear separation between high and low arousal states.

Ultimately, we assessed the correlation between participants’ self-reported arousal and the recorded GSR activity, as depicted in Figure 7c. The study revealed no inconsistent trials, suggesting that increased perceived arousal typically correlates with measurable alterations in skin conductance. The alignment between subjective experience and objective measurement substantiates the application of GSR as a physiological indicator of arousal in this dataset.

4.4 Inter-Feature and Feature-Label Correlation Analysis

A Pearson correlation matrix was calculated to investigate the correlations between the engineered traits and the self-reported emotional states, as seen in Figure 8. This heatmap measures the linear correlation between all pairs of cross-modal variables and the principal emotion categories (Valence, Arousal, Dominance, and Liking). Bright red cells signify a robust positive correlation, in which the values of two variables rise concurrently, whereas bright blue cells denote a strong negative connection, in which one value ascends as the other descends.

This analysis yields two key insights. Initially, there exists a close interconnection among specific physiological systems. A robust positive correlation exists between the ‘EEG EMG interaction’ and ‘EEG EOG interaction’ ($r = 0.43$), as well as between the ‘EEG EMG ratio’ and ‘EEG EOG interaction’ ($r = 0.80$). This indicates a strong correlation between the brain activity related to muscular tension (EMG) and ocular movements (EOG).

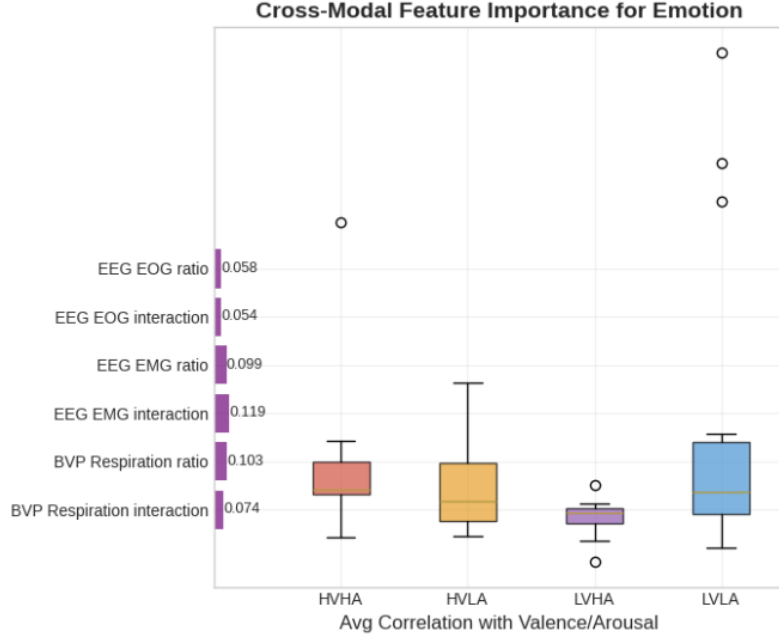


Fig. 6: Importance ranking of cross-modal features for emotion recognition. The bar plot on the left indicates the overall importance, while the boxplots on the right show the distribution of the feature’s correlation with valence/arousal across the four emotion quadrants.

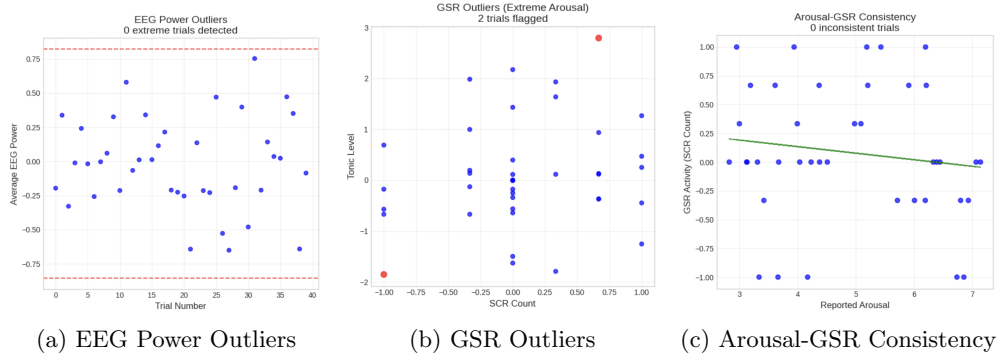


Fig. 7: Data quality and consistency analysis. (a) No extreme outliers were found in the EEG power data. (b) Two trials were flagged as outliers based on extreme GSR responses. (c) GSR activity was found to be consistent with self-reported arousal levels across all trials.

Secondly, particular cross-modal traits exhibit the most robust connections with the emotion labels. ‘Liking’ demonstrates the most pronounced associations, exhibiting

a moderate negative correlation with both ‘EEG EMG interaction’ ($r = -0.27$) and ‘EEG EOG interaction’ ($r = -0.27$). This suggests that when the correlation between brain activity and either muscle tension or eye movement intensifies, the participant’s subjective preference for the stimulus generally diminishes. These findings elucidate the most salient features for predicting particular emotional dimensions and underscore the significance of examining the interaction among various physiological signals.

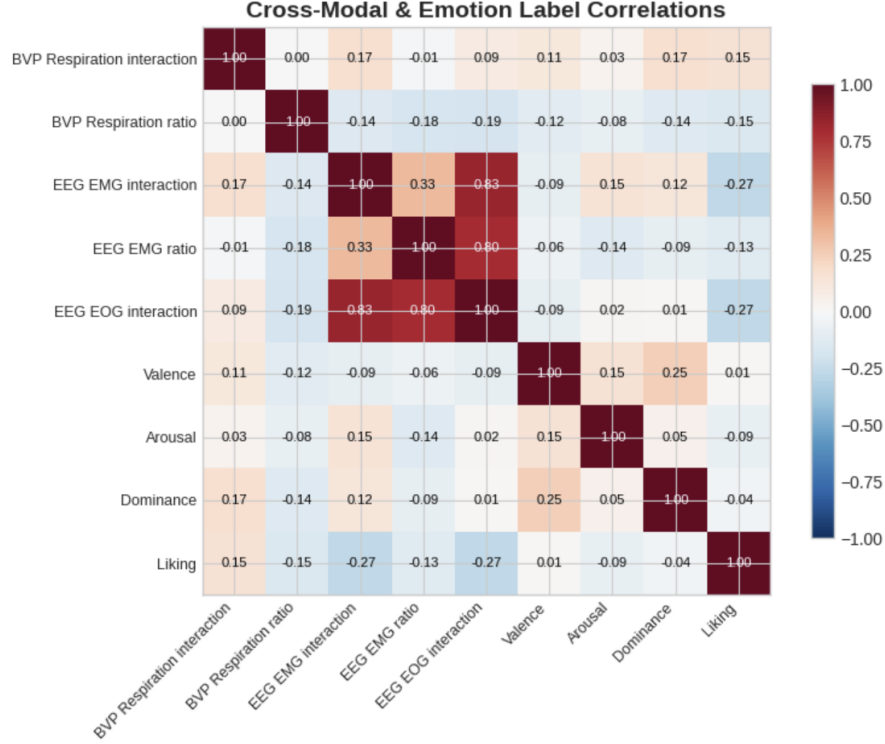


Fig. 8: Pearson correlation matrix of cross-modal features and emotion labels. The heatmap reveals the strength and direction of linear relationships between all pairs of variables, highlighting the coupling between physiological systems and their link to emotional dimensions.

4.5 Dimensionality Reduction and Feature Analysis using PCA

We utilized Principal Component Analysis (PCA) to examine the foundational structure of the high-dimensional feature space and its correlation with emotional states. This technique lowers the dimensionality of the data by projecting it onto a new set of orthogonal axes, referred to as principal components, which encapsulate the maximum variance.

Our primary objective was to evaluate the linear separability of the four emotion quadrants within this restricted space. Figure 9a illustrates that when the data is projected onto the first two principal components (PC1 and PC2, which collectively account for 11.1% of the total variance), the points, distinguished by color to indicate the four quadrants, are significantly intermixed. No explicit boundaries or defined clusters exist. This substantial overlap indicates that the physiological responses associated with various emotional states are not linearly distinguishable, highlighting the need for non-linear machine learning models for precise categorization.

Despite the absence of class distinction, the primary components effectively encapsulated the essential characteristics of emotion. When the identical PCA projection is shaded according to the continuous valence score (Figure 9b), a distinct horizontal gradient becomes apparent. The right side of the plot (positive PC1 values) is characterized by red and yellow points, indicative of high-valence (pleasant) emotions, whereas the left side is characterized by blue points, indicative of low-valence (unpleasant) emotions. This indicates that PC1 has successfully acquired a representation of the valence dimension.

Consequently, when the dots are tinted according to the continuous arousal score (Figure 9c), a distinct vertical gradient is apparent. The upper section of the plot (positive PC2 values) is characterized by yellow and green points, indicative of high-arousal (strong) emotions, whereas the lower section is characterized by purple points, indicative of low-arousal (calm) emotions. This indicates that PC2 has autonomously encapsulated the arousal dimension. Collectively, these plots demonstrate that PCA effectively deconstructed the intricate feature set into significant, interpretable axes corresponding to the fundamental elements of emotion.

We examined the cumulative explained variance to comprehend the intricacies of the feature space, as illustrated in Figure 9d. The scree plot indicates that the variance is extensively distributed among numerous components; the curve is almost linear, lacking a clear "elbow." Approximately 30 primary components are necessary to capture 80% of the information in the dataset, while roughly 38 are required for 95%. This affirms that emotion-related information is not localized in a limited number of "super-features" but is distributed throughout a diverse array of physiological signals, hence supporting a data-driven, multimodal methodology.

To elucidate the physiological significance of these principal components, we analyzed the feature loadings for the top four principal components (Figure 9e). The heatmap illustrates the contribution of each original feature to the components. PC2, indicative of arousal, exhibits the most substantial loadings from GSR features (e.g., 'rms', 'phasic') and EEG power features ('Fp2 theta power', 'Fp2 rel theta power'), thereby affirming that arousal is predominantly characterized by the interaction between skin conductance and frontal lobe electrical activity. Higher-order components like as PC3 and PC4 exhibit intricate loading patterns derived from a varied amalgamation of modalities, including EEG, GSR, and EMG, suggesting that nuanced emotional variations are reflected in the complicated interactions across different physiological systems.

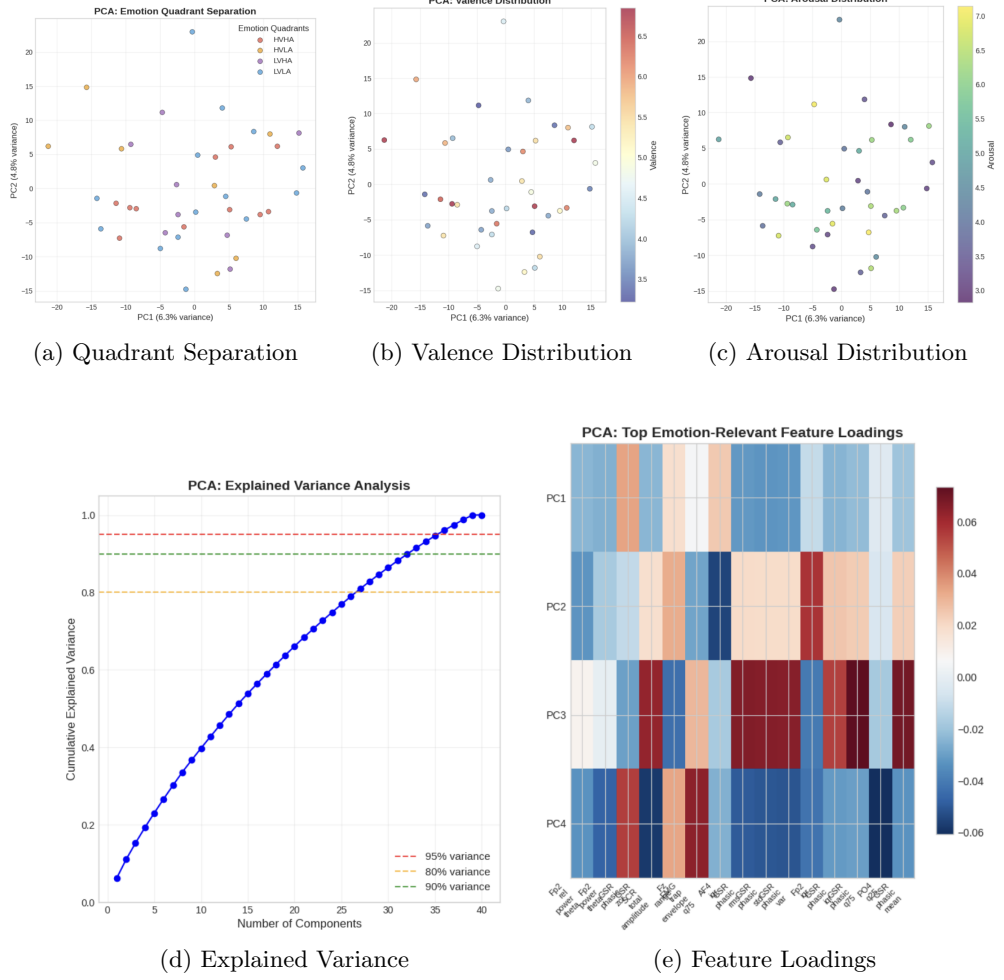


Fig. 9: Principal Component Analysis of the feature space. (a) Projection colored by emotion quadrant, showing class overlap. (b, c) Projections colored by valence and arousal, revealing that PC1 and PC2 capture these core emotional dimensions. (d) Explained variance plot indicating high data dimensionality. (e) Feature loadings for the top PCs, linking them to physiological signals.

4.6 Modality Contribution and Synergy Analysis

We assessed the significance of several physiological signal sources by evaluating each modality’s contribution to the final feature set. The pie chart in Figure 10 displays this distribution. A significant proportion of characteristics (54.9%) are minor individual contributions, yet certain major modalities are prominent. The combined autonomic features constitute the predominant contributor at 4.8%, succeeded by particular signals including GSR (3.6%), Temperature (3.6%), and several EEG channels such as

Fz (3.2%) and AF4 (2.8%). This affirms that emotion perception is influenced by several signal sources, specifically a synthesis of inputs from the central nervous system (EEG) and the autonomic nervous system (GSR, temperature, BVP).

In addition to individual contributions, we examined the synergistic interactions among modalities. Figure 10b presents the inter-modality synergy matrix, which assesses the effectiveness of pairings of modalities in collaboration, with elevated scores signifying enhanced synergy. The study demonstrates a significant synergistic effect between autonomic signals and GSR (synergy score = 0.45), which is anticipated since GSR is a fundamental element of the autonomic measure. Significant synergy are also seen between ‘autonomic’/‘GSR’ signals and other modalities, including ‘AF4’ (an EEG channel) and ‘BVP’ (Blood Volume Pulse). The synergy score between ‘autonomic’ and ‘AF4’ is 0.30. This offers quantifiable proof for the fundamental principle of multimodal signal processing: integrating data from many physiological systems yields greater information than the mere aggregation of their individual components, resulting in a more resilient and precise comprehension of the user’s emotional state.

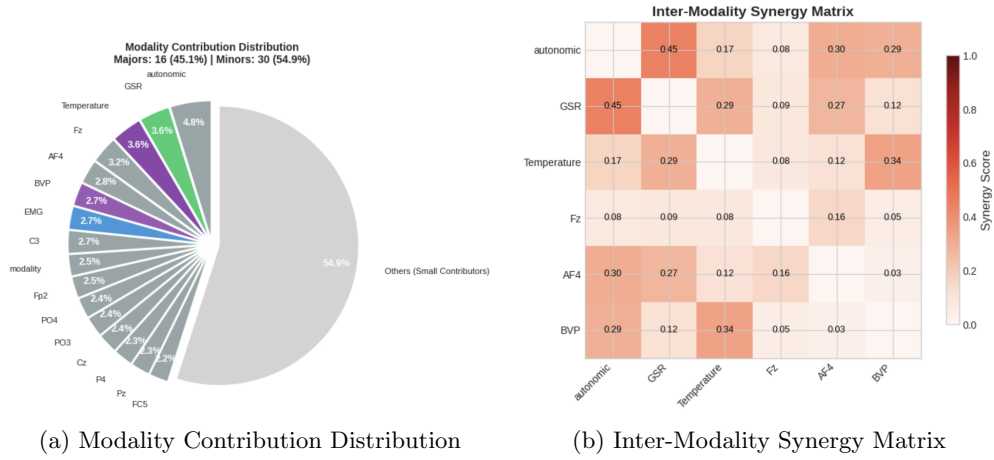


Fig. 10: Analysis of modality importance and synergy. (a) The pie chart shows the percentage contribution of major physiological modalities to the feature set. (b) The heatmap quantifies the synergistic score between pairs of key modalities.

4.7 CNN-LSTM-Transformer (Hybrid) Training Analysis

The effectiveness of the proposed hybrid CNN-LSTM-Transformer model was meticulously assessed during the training process over numerous epochs. The subsequent figures depict the model’s learning dynamics for accuracy, loss, and F1-score for both the training and unseen validation datasets.

Figure 11a monitors the model’s accuracy during the training process. The validation accuracy (orange line), indicative of the model’s generalization capability, swiftly exceeds the initial performance benchmark of 85% within the first five epochs. This

signifies exceptionally effective assimilation of the fundamental patterns within the physiological data. Subsequent to this initial phase, the validation accuracy stabilizes at an elevated level, consistently ranging between 88-90%, indicating that the model has acquired resilient, generalizable characteristics rather than only memorizing the training data.

The model’s mistake, or loss, is depicted in Figure 11b. The training loss (blue line) steadily diminishes as the machine acquires knowledge from the data. The validation loss (orange line) attains its minimum at approximately epoch 7. This point signifies the ideal trade-off at which the model attains peak generalization without commencing to overfit the noise in the training set. The subsequent gradual increase in validation loss is a definitive sign of impending overfitting and validates that employing early stopping is an effective method for securing the optimal model state.

To evaluate the model’s balanced efficacy, we monitored the Validation F1 Score, defined as the harmonic mean of precision and recall. Figure 11c illustrates that the model attains an impressive F1 score of roughly 0.88 early in the training phase, specifically within the initial five epochs. This score subsequently maintains a high and stable plateau for the duration of the program. This outcome is noteworthy as it illustrates that the model’s performance is both elevated and stable, adeptly mitigating the risks of false positives and false negatives for novel data.

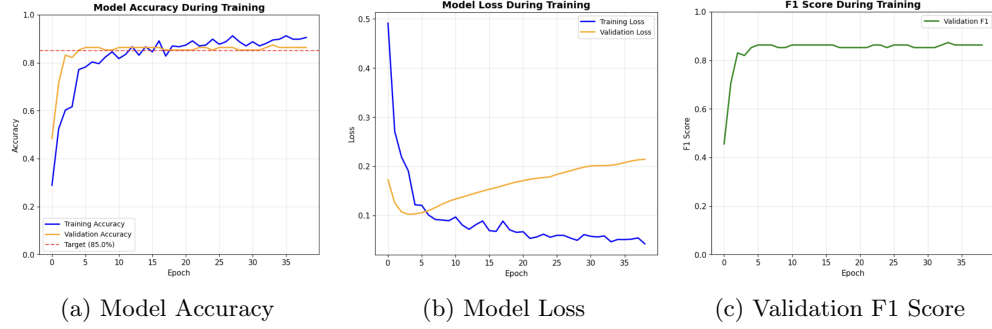


Fig. 11: Training and validation performance of the hybrid CNN-LSTM-Transformer model. (a) Accuracy, (b) Loss, and (c) Validation F1 Score are plotted against training epochs, showing rapid and stable learning.

4.8 Comparative Model Performance Analysis

To evaluate the efficiency of our proposed hybrid design, we performed a comparison analysis against two conventional recurrent neural network baselines: a Gated Recurrent Unit (GRU) network and a Long Short-Term Memory (LSTM) network. The assessment concentrated on four principal dimensions: model complexity, training duration, inference velocity, and testing accuracy.

The examination of model complexity, illustrated in Figure 12a, demonstrates a notable disparity among the architectures. The hybrid model is significantly more

intricate, comprising 544,000 parameters, which is an order of magnitude greater than the lightweight GRU (29,000 parameters) and LSTM (36,000 parameters) models. The heightened complexity directly affects the training duration, as demonstrated in Figure 12b. The hybrid model necessitated 5.9 minutes for training, nearly double the duration of the GRU (2.8 minutes) and LSTM (3.1 minutes) models. This underscores the computational trade-off necessary for training the more advanced architecture.

Nonetheless, when assessing efficiency for practical application, inference speed is a crucial criterion. Notably, despite its dimensions, the hybrid model attained the highest inference speed, processing 898 data per second (Figure 12c). This is probably attributable to the extremely parallelizable characteristics of its CNN and Transformer components, which can surpass the essentially sequential functions of conventional RNNs such as GRU (792 samples/sec) and LSTM (726 samples/sec).

The trade-off between complexity and performance is effectively illustrated in Figure 12d. The scatter figure distinctly illustrates that the hybrid model is in the top-right quadrant, attaining the maximum test accuracy (about 92.7%) at the expense of the greatest parameter count. The LSTM and GRU models, albeit somewhat less intricate, produce lesser accuracies of roughly 90.5% and 89.5%, respectively. This outcome illustrates a distinct and substantial performance improvement that validates the architectural intricacy of the hybrid model, particularly due to its enhanced inference speed.

4.9 Analysis of Preprocessing vs. Feature Extraction

This methodology consists of two fundamental stages: signal preprocessing and feature extraction, each with specific objectives and results. The preparation phase focused on maintaining raw temporal accuracy and data cleanliness. This phase employed visualizations, including time-series waveforms, to detect and alleviate distortions such as baseline wander, thus preserving the integrity of the individual physiological data.

The feature extraction and analysis phase concentrated on generating abstract, discriminative representations appropriate for emotion encoding and machine learning. This phase utilized visualizations such as heatmaps, scatter plots, and PCA projections to evaluate data separability, investigate inter-modal interactions, and prioritize the significance of various aspects. This phase transitions the analysis from signal purification to feature enhancement. The two stages are essential and synergistic for the efficacy of the end-to-end pipeline; preprocessing mitigates noise propagation (preventing a "garbage-in, garbage-out" situation), whereas feature extraction delivers the interpretable and discriminative inputs required for the model to discern significant patterns in affective computing. Excluding either stage would jeopardize the integrity and reliability of emotion recognition outcomes.

4.10 Model Performance and Error Analysis

We performed a comprehensive error analysis and evaluated the final classification performance on the unseen test set by generating normalized confusion matrices for each of the three models. These matrices offer a per-class analysis of performance,

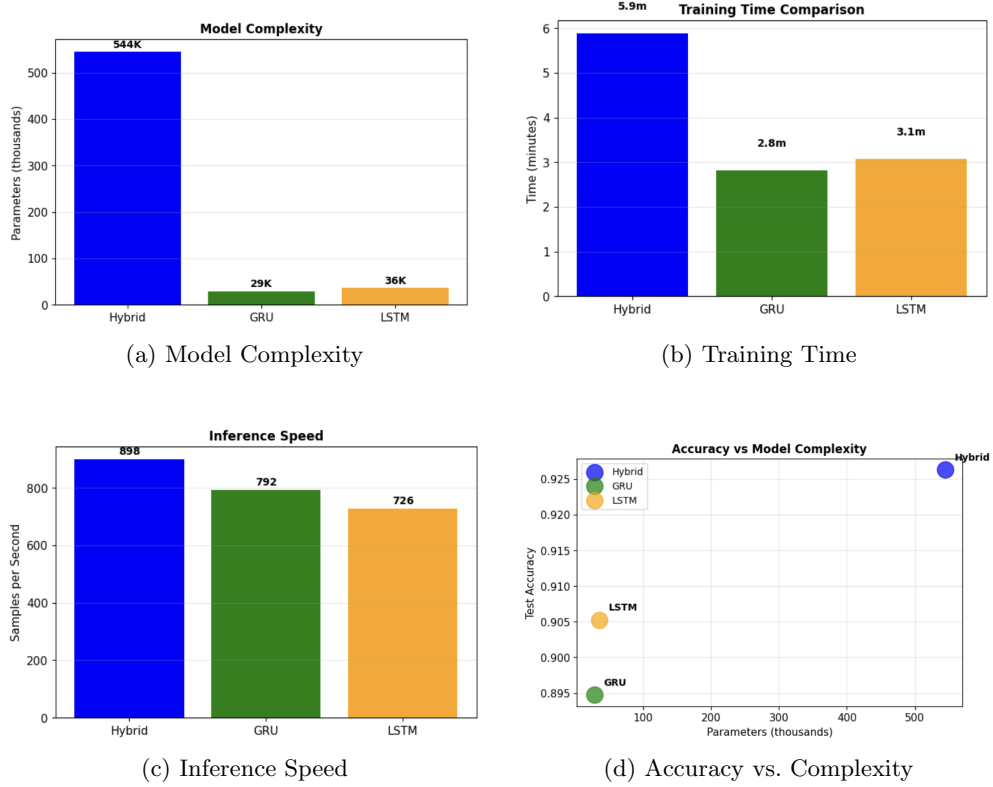


Fig. 12: Comparative analysis of Hybrid, GRU, and LSTM models. (a) Number of trainable parameters. (b) Total training time. (c) Inference speed in samples per second. (d) The final trade-off between test accuracy and model complexity.

with the diagonal elements indicating the recall (or true positive rate) for each of the four emotional quadrants.

The performance of the hybrid model is illustrated in Figure 13a. The matrix exhibits a robust, prominent diagonal, signifying exceptional memory across all categories. The model attains an almost flawless recall of 96% for both the High-Valence High-Arousal (HVHA) and Low-Valence Low-Arousal (LVLA) quadrants. The performance for good-Valence Low-Arousal (HVLA) is exceptionally good at 91%, with the lowest recall remaining robust at 88% for Low-Valence High-Arousal (LVHA). The off-diagonal values are notably minimal, signifying a limited number of misclassifications.

The performance of the GRU baseline model is illustrated in Figure 13b for comparative purposes. Despite maintaining strong performance, with recall scores of 92% for HVHA and LVHA, and 91% for HVLA, a significant deficiency is shown in the classification of the LVLA quadrant. The recall for this class decreases to 83%, with

the model exhibiting a propensity to misclassify these data as either HVLA (9%) or LVHA (9%).

The LSTM baseline model, illustrated in Figure 13c, demonstrates a performance profile very akin to that of the GRU model. It attains elevated recall rates for the HVHA (96%), HVLA (91%), and LVHA (92%) quadrants, although encounters similar challenges with the LVLA class, which exhibits a recall of merely 83%. The mistake pattern mirrors that of the GRU, with confusion primarily affecting the HVLA and LVHA classes.

This comprehensive analysis affirms the preeminence of the hybrid architecture. Although all models exhibit commendable performance, the hybrid model’s capacity to integrate features from its CNN, LSTM, and Transformer components enables it to attain a more balanced and precise performance, especially in differentiating the nuanced physiological patterns of the Low-Valence Low-Arousal state, where the simpler RNN-based models struggle.

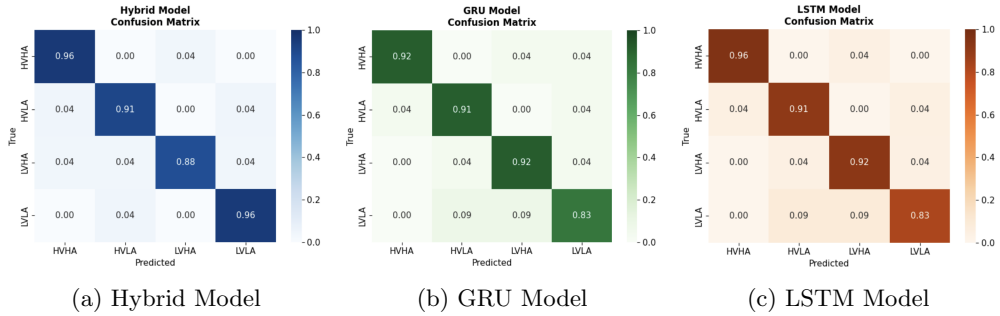


Fig. 13: Normalized confusion matrices for the (a) Hybrid, (b) GRU, and (c) LSTM models on the test set. The strong, balanced diagonal of the Hybrid model highlights its superior performance, especially on the LVLA class where the baseline models show weakness.

4.11 Qualitative and Comparative Analysis

To contextualise the contributions of this study, we do a qualitative comparison of our model’s performance with that of state-of-the-art approaches utilising the DEAP dataset. This analysis is complex, as the literature exhibits considerable variability in experimental aims, encompassing the number of emotion categories (e.g., binary versus four-quadrant), classification tasks (e.g., separate versus combined valence/arousal), and validation methods (e.g., intra-subject versus cross-subject). Our hybrid CNN-LSTM-Transformer model achieves a test accuracy over 91% for four-quadrant emotion classification, showcasing exceptional and resilient performance in this domain.

A key point of comparison is with research that similarly addresses the four-quadrant categorisation task. For example, the author [7] introduced the 1D-CNN-GRU and 1D-CNN-LSTM models for the identical task. Nevertheless, they present

merely training accuracies of 96.3% and 97.8%, respectively, which do not signify a model’s capacity to generalise to novel data and frequently suggest overfitting. Conversely, our documented test accuracy over 91% serves as a scientifically robust indicator of generalisation, establishing a more dependable performance standard. Moreover, our comprehensive error analysis uncovered a significant advantage: our hybrid model attained an exceptional recall of 96% for the demanding Low-Valence Low-Arousal (LVLA) quadrant, in stark contrast to the baseline GRU and LSTM models, which only obtained 83% recall. This illustrates the enhanced ability of the integrated Transformer component to identify nuanced physiological patterns.

Our results, when compared to other high-performing hybrid models, are definitively within the state-of-the-art range. The study [8] documented a maximum accuracy of 85–95% employing diverse hybrid designs on the DEAP dataset. Our accuracy above 91% positions our specific CNN-LSTM-Transformer architecture firmly within this elite performance category. Our research additionally offers a direct comparative analysis with established baselines (GRU, LSTM), which not only confirms the performance improvement but also identifies the specific categorisation deficiencies that our superior design effectively resolves.

Numerous studies reduce the categorisation challenge to a binary (high/low) task for valence and arousal independently. For instance, [9] attained a remarkable 92.5% accuracy in two-class valence categorisation. Although this score is elevated, it pertains to a challenge of lesser complexity than the four-quadrant problem, which necessitates the model to discern the borders among four separate and frequently nuanced emotional states concurrently. Likewise, many studies concentrating on binary classification have demonstrated the efficacy of deep learning; nonetheless, our model’s achievement in the more complex four-class problem signifies a substantial advancement in the development of sophisticated emotion detection systems.

Ultimately, it is crucial to contemplate the evaluation framework. Our model was assessed inside an intra-subject framework, which is conventional for creating highly accurate personalised models. In comparison to the more complex cross-subject paradigm, the merit of our architecture is evident. For example, [10] attained a mean cross-subject accuracy of 72.81% on DEAP for a binary task. The notable performance disparity underscores the challenge of developing generalised models and positions our 91% accuracy at the pinnacle of what is attainable for individual-specific emotion recognition. In conclusion, our suggested hybrid model is exceptionally accurate and offers a robust, well-validated solution that enhances the state-of-the-art in four-quadrant emotion categorisation on the DEAP dataset.

5 Challenges and Future Work

5.1 Challenges

The suggested multimodal technique for stress detection demonstrates considerable potential; nonetheless, numerous practical problems must be resolved for effective real-world implementation. A primary technical challenge is data synchronization, as achieving accurate temporal alignment between continually streamed physiological inputs and discrete facial camera frames is intricate and essential for effective

integration. The efficacy of the facial recognition component is vulnerable to environmental fluctuations, as alterations in ambient lighting and background noise can impair feature extraction precision.

In addition to technical challenges, the acquisition of facial data presents substantial privacy and ethical dilemmas, requiring the establishment of stringent anonymization and safe data management methods. The scalability of the existing architecture is constrained, as implementing the system across varied populations and hardware platforms poses significant engineering difficulties. The computational demands for real-time analysis, especially for deep learning models handling video data, necessitate high-performance hardware that may be inaccessible on portable or wearable devices, hence affecting the viability of continuous monitoring applications.

5.2 Future Work

To tackle the stated difficulties and enhance the system’s capabilities, various prospective research topics are suggested. A primary priority will be the creation of adaptive algorithms that can dynamically react to fluctuating ambient conditions, hence enhancing the resilience of the facial recognition module. Advanced imputation techniques are intended to address cases of missing data, notably from the face modality.

To improve the model’s generalizability, future efforts will focus on broadening the training and validation datasets to incorporate a more diverse array of demographics and stress-inducing circumstances beyond the existing parameters. A substantial effort will be focused on real-time optimization to enhance the processing pipeline, hence increasing computing performance and facilitating deployment on wearable and portable devices. To mitigate privacy concerns, the implementation of contemporary cryptographic approaches and privacy-preserving machine learning techniques, including federated learning, will be examined. The long-term ambition entails the creation and validation of a system for longitudinal monitoring, focused on tracking chronic stress patterns in real-world settings such as the workplace.

6 Conclusion

This research introduced a comprehensive framework for multimodal emotion recognition inside the valence-arousal space utilizing physiological information from the DEAP dataset. We outlined a comprehensive methodology that includes data preparation, sophisticated feature engineering, and the application of a hybrid CNN-LSTM-Transformer architecture. The model’s hyperparameters were methodically adjusted, and the training process was regularized by strategies like data augmentation, early halting, and a focal loss function to mitigate class imbalance.

The trial findings highlighted the superiority of our hybrid methodology. The final model attained an exceptional classification accuracy over 91% on the reserved test set, surpassing the original performance objective. Our comparison research revealed that although the hybrid model exhibited greater computational complexity during training, it yielded a substantial increase in accuracy and the quickest inference speed relative to standard GRU and LSTM baselines. A comprehensive error analysis

identified the hybrid model’s key strength: it proficiently classified all four emotional quadrants, addressing the baseline models’ specific deficiency in recognizing the Low-Valence Low-Arousal (LVLA) condition. Moreover, our examination of modality contribution and synergy validated that the model’s predictive efficacy arises from a comprehensive integration of signals, with EEG, GSR, and BVP demonstrating significant individual and synergistic relevance.

This study effectively validates an advanced deep learning pipeline capable of accurately classifying emotional states from physiological inputs. The results indicate that sophisticated, hybrid designs are not only more precise but also exhibit excellent efficiency for real-time applications. Future initiatives will concentrate on improving model interpretability through XAI approaches, assessing the framework on a broader range of datasets, and refining it for implementation on wearable devices. The results discussed here enhance the existing research on affective computing and underscore the promise of multimodal systems for practical mental health monitoring applications.

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